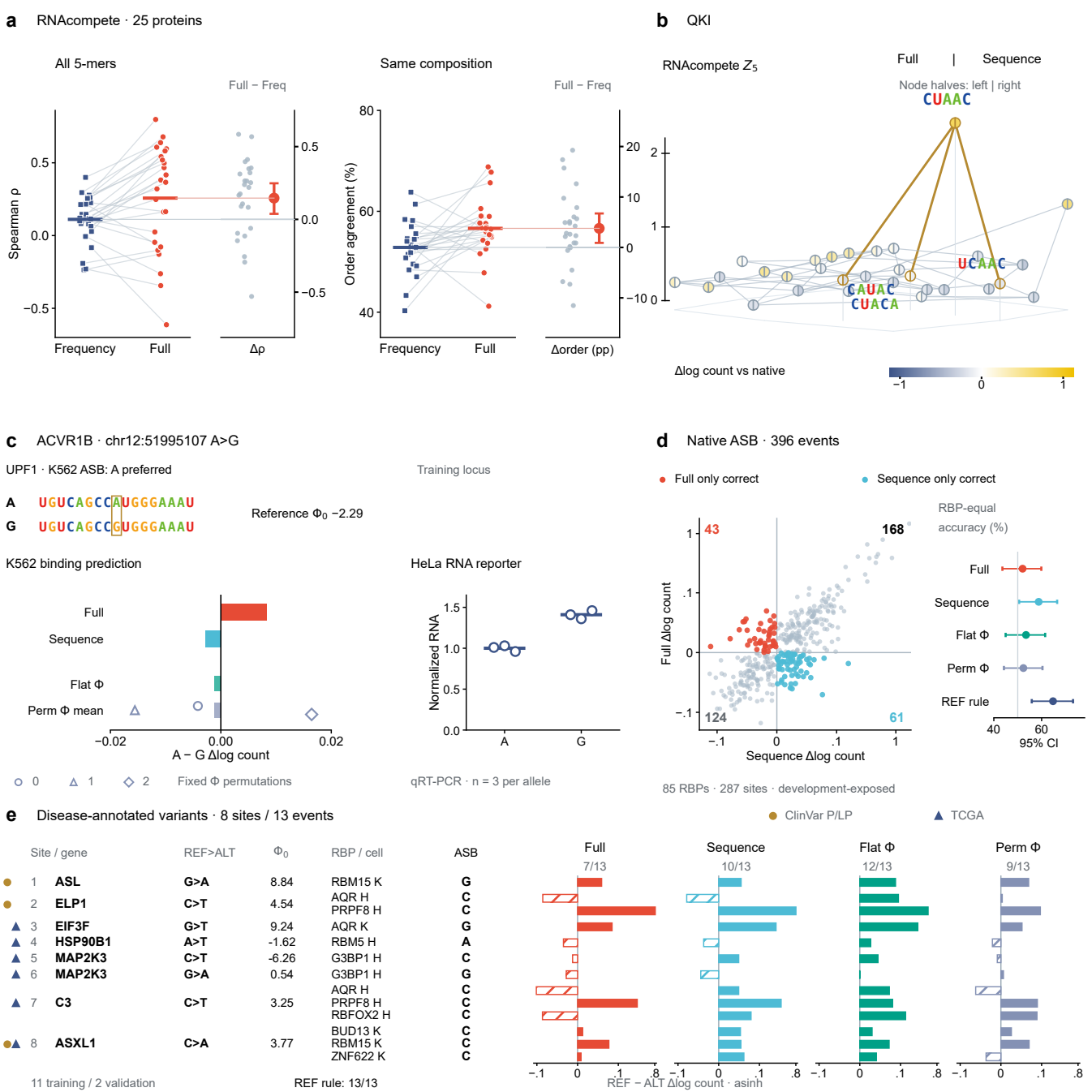

a
RNAcompete · 25 proteins
All 5-mers
Full − Freq
Spearman ρ
Frequency
Full
Δρ
Same composition
Full − Freq
Order agreement (%)
Frequency
Full
Δorder (pp)
b
QKI
RNAcompete Z_5
Full | Sequence
Node halves: left | right
CUAAC
UCAAC
CAUAC
CUACA
Δlog count vs native
c
ACVR1B · chr12:51995107 A>G
UPF1 · K562 ASB: A preferred
Training locus
A UGUCAGCCAUGGGAAAU
G UGUCAGCCGUGGGAAAU
Reference Φ0 −2.29
K562 binding prediction
Full
Sequence
Flat Φ
Perm Φ mean
A − G Δlog count
Fixed Φ permutations
HeLa RNA reporter
Normalized RNA
qRT-PCR · n = 3 per allele
d
Native ASB · 396 events
Full only correct
Sequence only correct
43
168
124
61
Full Δlog count
Sequence Δlog count
RBP-equal accuracy (%)
Full
Sequence
Flat Φ
Perm Φ
REF rule
95% CI
85 RBPs · 287 sites · development-exposed
e
Disease-annotated variants · 8 sites / 13 events
ClinVar P/LP
TCGA
Site / gene
REF>ALT
Φ0
RBP / cell
ASB
1 ASL G>A 8.84 RBM15 K G
2 ELP1 C>T 4.54 AQR H C
PRPF8 H C
3 EIF3F G>T 9.24 AQR K G
4 HSP90B1 A>T -1.62 RBM5 H A
5 MAP2K3 C>T -6.26 G3BP1 H C
6 MAP2K3 G>A 0.54 G3BP1 H G
7 C3 C>T 3.25 AQR H C
PRPF8 H C
RBFOX2 H C
8 ASXL1 C>A 3.77 BUD13 K C
RBM15 K C
ZNF622 K C
11 training / 2 validation
REF rule: 13/13
Full 7/13
Sequence 10/13
Flat Φ 12/13
Perm Φ 9/13
REF − ALT Δlog count · asinh